

Abstract

Scientists discovered a triply-lensed object by observing galaxy cluster PLCK G165.7+67.0 through the Prime Extragalactic Areas for Reionization and Lensing Science (PEARLS) project of the James Webb Space Telescope (JWST.) Through additional studies on the gravitationally lensed object, including the use of the Director’s Discretionary Time of JWST, the object was discovered to be a Type Ia Supernova, which was named Supernova H0pe. Scientists analyzed the relative time-delay distances between images of the SN and compared those values to different lens models that corresponded to a certain Hubble constant. Through the chosen models, scientists were able to identify a plausible H_0 value, around $H_0 = 75.7 - 5.5 + 8.1$ km/s/Mpc. This is the first Hubble constant precision measurement from a multiply-imaged SN Ia without use of cosmic ladder or CMB methodology.

Introduction

The Hubble tension is one of the most disputed topics in astronomy, where two methods of measuring the constant of the rate of expansion of the universe (the Hubble constant, or H_0) yield values that do not agree with each other. One of these methods looks at the cosmic microwave background to measure the rate of the expansion of the universe in its infancy. Through the standard model of cosmology (Lambda CDM), they project this rate forwards, resulting in a H_0 constant of around 67-68 km/s/mpc. The second method utilizes a ‘cosmic ladder.’ By using standard candles such as Cepheid variable stars and Type Ia supernovae, we can calculate their distance from the Earth, beginning locally, then extending into the universe. This method yields a H_0 value of around 70-76 km/s/mpc

Type Ia Supernovae (or SN Ia) are supernovae that originate from binary systems made up of a main sequence star and a white dwarf. This white dwarf is a ‘husk’, a star that has exhausted its fuel for nuclear fusion, and is now composed of electron degenerate matter, carbon gas, and oxygen gas. If the two stars are close enough, the white dwarf acts a ‘cosmic vampire,’ accreting the main sequence star’s material onto itself. This material builds up on the white dwarf, increasing the pressure and temperature within the white dwarf’s core. As the mass creeps towards 1.44 solar masses (a specific mass known as the Chandrasekhar limit), the star begins to contract. However, in a degenerate star, it cannot expand to cool down and counteract the rising temperature. As the mass of the star begins to reach 1.44 solar masses, the star explodes. Due to the common mass at which every Ia SN explodes, their explosions have almost exactly the same absolute magnitude (including some measurement error.) This explains why Ia SN are considered ‘standard candles,’ making them useful for making cosmic distance measurements in astronomy due to their predictability and consistent luminosities. In addition, Ia SN have distinct silicon absorption lines as a product of carbon fusion and a lack of strong hydrogen and helium absorption lines, due to its core being made of primarily carbon and oxygen as a white dwarf. This makes a Type Ia supernova relatively easy to identify and use.

Gravitational Lensing: according to the theory of general relativity, gravitational lensing is a phenomenon where objects with mass can warp the fabric of spacetime around it. The greater the mass of the object, the more warping occurs. When light from a distant source passes by a massive foreground object (such as a black hole or a large galaxy cluster), the path of the light bends around the object from the warping of spacetime. Because there are multiple paths that the light can take, the observer may see multiple images of the distant object. The object can also be magnified or distorted (**see image 1 and 3**)— hence the name ‘gravitational lensing.’

Supernova H0pe: an attempt to solve the Hubble tension

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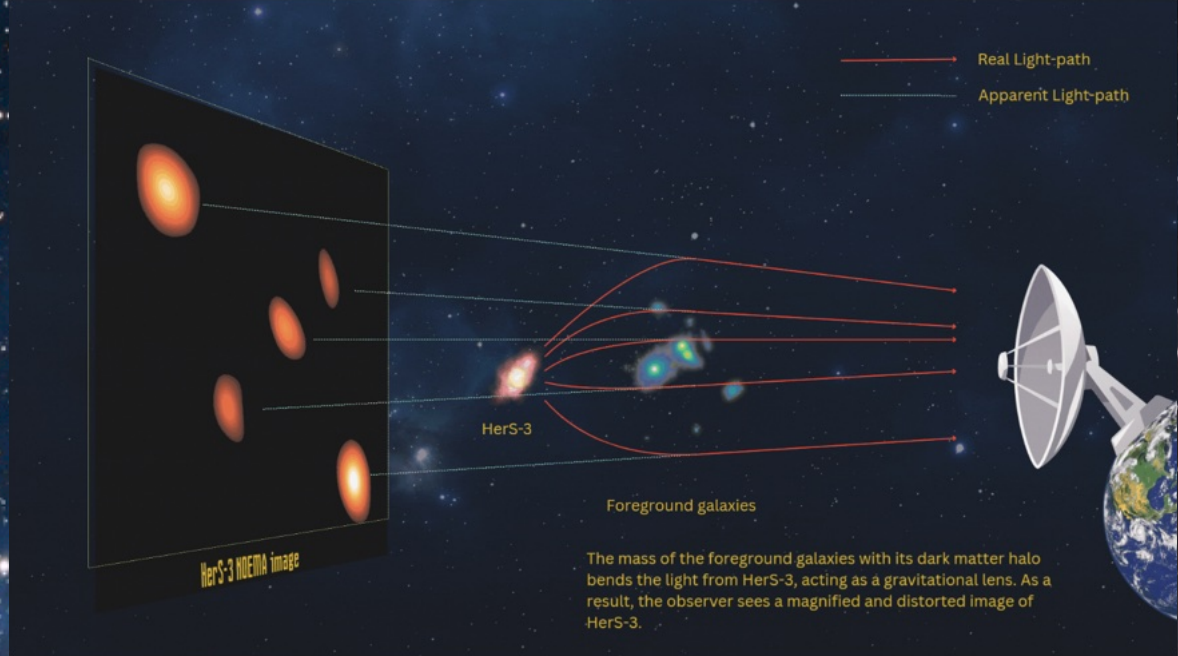


Image 1. An illustration of gravitational lensing. Graphic showing the arrangement that leads to the formation of an Einstein Cross (Credit : N. Lira, P. Cox et al / NOEMA / ALMA (ESO/NAOJ/ NRAO)

Introduction (cont.)

Time delay in gravitational lensing is when multiply lensed images of a distant source object arrive at the observer at different times. This is cause by two phenomena: geometric delay and Shapiro delay. When geometric delay happens, the physical paths of the light differ in length. Geometrically, the light taking a longer path will arrive at a later time. Shapiro delay is a relativistic effect where the warping of spacetime acts as ‘speed bump.’ As light nears the massive object or cluster, it takes a longer time to travel when passing through the denser gravitational field of the object.

Methods

The discovery of Supernova H0pe occurred through the Prime Extragalactic Areas for Reionization and Lensing Science (PEARLS) project. Their main goals are to study the epoch of galaxy assembly, AGN growth, and first light using the James Webb Space Telescope. When imaging PLCK G165.7+67.0, scientists noticed a multiply lensed object near the galaxy cluster. In order to explore SN H0pe further, the PEARLS-Clusters team obtained JWST’s DDT (Director's Discretionary Time) observations, through which they confirmed that SN H0pe was a type Ia SN due to its blue shifted silicon spectra (**see image 2.**) In addition to the JWST, data was acquired on the MMT on Mt. Hopkins, and at the Large Binocular Telescope on Mt. Graham, both in Arizona. By analyzing both observations, the team was able to confirm that SN H0pe was anchored to a background galaxy, well behind G165.

JWST’s NIRCam was used to measure UV-near-infrared wavelengths of the supernova, due to the high sensitivity of a supernova to UV wavelengths during explosion. The UV wavelengths become redshifted ($z = 1.7833 \pm 0.0005$) into the infrared spectrum. Astronomers then took photos every three observer weeks, or one rest-frame week. Each photo contained three gravitationally-lensed instances of the supernova through seven different filters. In total, there were 9 different images of the SN explosion. This combined data set through different filters was able to produce a final SN Ia light curve with 9 epochs and rest-frame UV–near-IR wavelength coverage.

Methods (cont.)

BayeSN, a statistical computer model, was used to create the SN Ia light curve mentioned above. Because a supernova changes colors as it ages (e.g. it might start in high-energy wavelengths and shift to redder wavelengths over time), using only one wavelength filter could not have created an accurate light curve. BayeSN was utilized to combine the spectra in seven different filters to build a light curve. It also accounted for the redshift value, $z = 1.7833$. The BayeSN model then applied microlensing curves to each image, where it accounted for smaller instances of gravitational lensing occurring from other astronomical objects between the observer and the source. After creating the final light curve of SN H0pe, the model inferred the best-fit time delays between the images.

Next, seven different models were constructed to map the foreground galaxy cluster. The BayeSN-calculated time delays were then compared with the lens maps, which offered varying H_0 values. An absolute magnification value was calculated by BayeSN, which was possible due to the ‘standard candle’ nature of SN H0pe. Astronomers then used this value to determine which model of the foreground galaxy cluster was correct— whichever model fit the absolute magnification value. The lens models that fit the absolute magnification value were chosen, and scientists used these models to calculate the final value of the Hubble constant.



Image 3. A colored photo of SN H0pe, resembling warm taffy. By NASA, ESA, CSA, STScI, B. Frye University of Arizona), R. Windhorst (Arizona State University), S. Cohen (Arizona State University), J. D'Silva (University of Western Australia, Perth), A. Koekemoer (Space Telescope Science Institute), J. Summers (Arizona State University),, CC BY 4.0, <https://commons.wikimedia.org/w/index.php?curid=153461043>

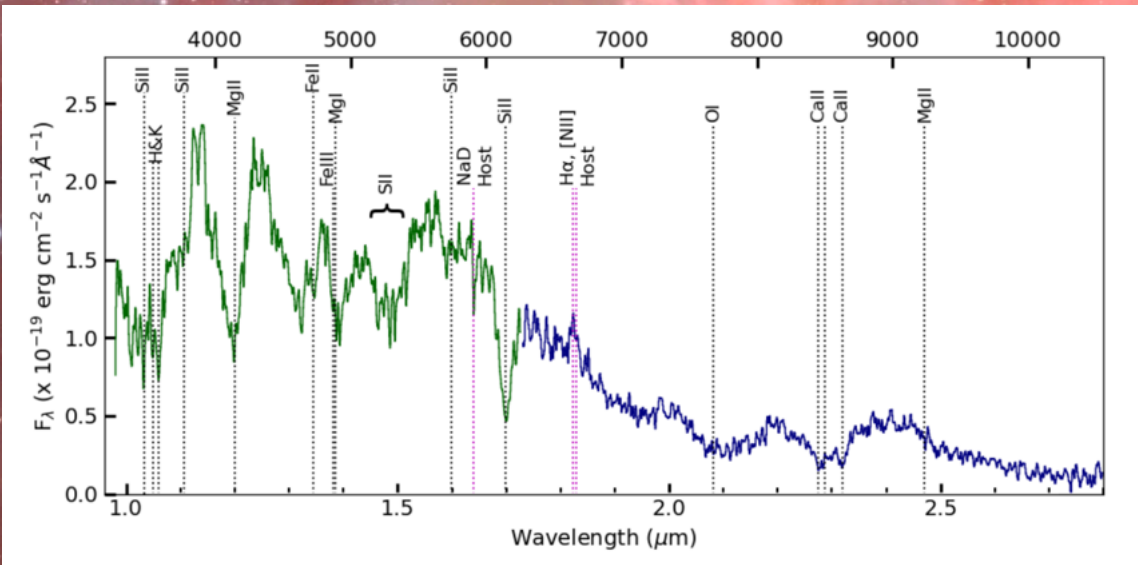


Image 2. Above is a graph detailing the high signal-to-noise NIRSpect spectrum of Supernova H0pe. Visible are the prominent line features, such as the [Si II] line blueshifted to rest frame ~ 6150 angstroms, a requisite line feature of a Type Ia supernova. A few features are also traced to the SN host galaxy, such as NaD, and H-alpha. The observed wavelength is given on the lower abscissa and the rest wavelength in the frame of the host galaxy is provided on the upper abscissa.

Results

From data collection by NIRCcam to time delay cosmography using BayeSN, astronomers arrived at the final value of $H_0 = 75.7 - 5.5 + 8.1$ km/s/mpc. This measurement is the first precision measurement of H_0 using a multiply-imaged type Ia supernova. Some feats within the methods used in obtaining this value include: the high signal-to-noise ratio of NIRSpect data, and the successful creation of a light curve through imaging nine epochs with seven filters using BayeSN.

Conclusion

Supernova H0pe exploded when the universe was just about 3.5 billion years old. Our calculated H_0 value was around 75.7 km/s/mpc. This value seems to agree with our method of the cosmic ladder, (70-76 km/s/mpc) as opposed to our calculated value using the Cosmic Microwave Background and Lambda CDM.

Within the Hubble tension, it was argued that local universe measurements have an error margin due to star crowding or blurry data. However, with a clear measurement using time-delays and standard candles, the Hubble constant calculated from SN H0pe refutes this argument. Thus, this deepens the Hubble tension and leads scientists to question the accuracy of our current model of the universe, Lambda CDM. Scientists now look to dark matter and dark energy for answers, in the case that an essential piece of physics is missing from our model of the universe.

Similar Phenomena

Einstein Rings

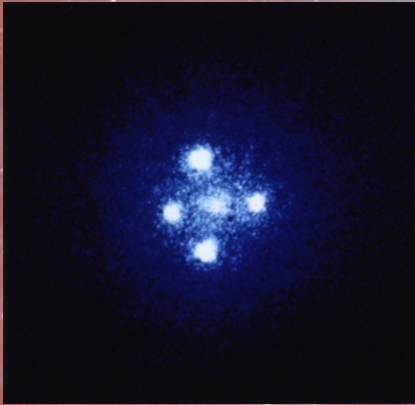
When observing a distant galaxy or object with a telescope, its light might encounter another galaxy on the way to the observer. The foreground galaxy acts as a magnifying lens due to its gravity, bending light rays. If the background light source and the gravitational lens are perfectly aligned, the image of the object appears as a ring. (**See image 4**)

Einstein Crosses

Einstein crosses also occur when light from a distant galaxy passes by a massive foreground object. This phenomena is called an Einstein cross because of his general theory of relativity, where he did not explicitly predict the multiply imaged object forming a cross, but rather the concept of gravitational lensing. Often, when the light source is perfectly aligned with the gravitational lens and Earth, the light source is multiply imaged in a cross shape (**see Image 5**)



(Image 4) Einstein ring, created by a nearer galaxy (NGC 6505) distorting light of a distant galaxy. Four images of the distant galaxy, corresponding to four bright spots, can be seen in this picture.



(Image 5) Einstein cross: Four images of the same distant quasar (due to the gravitational lensing of the galaxy closest to us, shown in the foreground, the Huchra Lens).

References

NASA, "Einstein cross: Four images of the same distant quasar." *Wikipedia*, Wikimedia Foundation, 13 Sept. 1990, en.wikipedia.org/wiki/Einstein_Cross#/media/File:Einstein_cross.jpg. Accessed 14 May 2026.

ESA, et al, "Einstein ring, created by a nearer galaxy (NGC 6505) distorting light of a distant galaxy" *Wikipedia*, Wikimedia Foundation, 10 Feb. 2025, en.wikipedia.org/wiki/Einstein_ring#/media/File:Closeup_of_the_Einstein_ring_around_galaxy_NGC_6505_ESA206346.jpg. Accessed 14 May 2026.